Assignment 3

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Part 1:

Below is the first part of code provided:

```
set.seed(67)
n <- 1000

sigma_A <- 25 # V_A
sigma.env <- 0

# Parent 1: Simulate effects of their two alleles
p1 <- rnorm(n,0,sqrt(sigma_A)/2) ##parental 1 allelic 1 effect
p2 <- rnorm(n,0,sqrt(sigma_A)/2) ##parental 2 allelic 2 effect

# Parent 2: Simulate effects of their two alleles
p3 <- rnorm(n,0,sqrt(sigma_A)/2) ##parental 1 allelic 1 effect
p4 <- rnorm(n,0,sqrt(sigma_A)/2) ##parental 2 allelic 2 effect


# Parent phenotypes:
p1.pheno <- p1 + p2  ##Parent 1: additive genetic effect
p2.pheno <- p3 + p4  ##Parent 1: additive genetic effect

# Add environmental noise:
p1.pheno <- p1.pheno + rnorm(n,0,sigma.env)
p2.pheno <- p2.pheno + rnorm(n,0,sigma.env)


# Start of offspring 1:
# Transmit allele from parent to offspring
which.allele.p1 <- rbinom(n, 1, 0.5) # transmitted
which.allele.p2 <- rbinom(n, 1, 0.5) # transmitted

# From parent 1
o1.1 <- rep(0, n) ## initialize
o1.1[which.allele.p1 == 0] <- p1[which.allele.p1 == 0]
o1.1[which.allele.p1 == 1] <- p2[which.allele.p1 == 1]

# From parent 2
o1.2 <- rep(0, n) ## initialize
o1.2[which.allele.p2 == 0] <- p3[which.allele.p2 == 0]
o1.2[which.allele.p2 == 1] <- p4[which.allele.p2 == 1]
```

```r
o1.pheno <- o1.1 + o1.2 + rnorm(n,0,sigma.env)

# Parent-offspring covariance
# From notes: V_A = 2*cov(x,z)

# Parent-offspring regression for H:
lmod <- lm(o1.pheno ~ p1.pheno)

h2.est <- 2 * coef(lmod)[2]
h2.est ## estimate for parent 1, just for illustration sake
```

```
##  p1.pheno
## 0.9521196
```

And now the code added for offspring 2:

```r
which.allele.p1 <- rbinom(n, 1, 0.5)
which.allele.p2 <- rbinom(n, 1, 0.5)

o2.1 <- rep(0, n)
o2.1[which.allele.p1 == 0] <- p1[which.allele.p1 == 0]
o2.1[which.allele.p1 == 1] <- p2[which.allele.p1 == 1]

o2.2 <- rep(0, n)
o2.2[which.allele.p2 == 0] <- p3[which.allele.p2 == 0]
o2.2[which.allele.p2 == 1] <- p4[which.allele.p2 == 1]

o2.pheno <- o2.1 + o2.2 + rnorm(n,0,sigma.env)
```

And the code for the monozygotic twins:

```r
## Twins
# Transmit allele from parent to offspring
which.allele.p1 <- rbinom(n, 1, 0.5) # transmitted
which.allele.p2 <- rbinom(n, 1, 0.5) # transmitted

# From parent 1
o1.1 <- rep(0, n) ## initialize
o1.1[which.allele.p1 == 0] <- p1[which.allele.p1 == 0]
o1.1[which.allele.p1 == 1] <- p3[which.allele.p1 == 1]

# From parent 2
o1.2 <- rep(0, n) ## initialize
o1.2[which.allele.p2 == 0] <- p3[which.allele.p2 == 0]
o1.2[which.allele.p2 == 1] <- p4[which.allele.p2 == 1]

o3.pheno <- o1.1 + o1.2 + rnorm(n,0,sigma.env)
o4.pheno <- o1.1 + o1.2 + rnorm(n,0,sigma.env)
```

Now for our confidence intervals of VsubA and h^2 from 100 100 and sigma.env set to 1 and 15:

Part 2:

```r
VA.est <- rep(0, 100) # 100 runs
h2.est <- rep(0, 100)

sigma.env <- 1

for(i in 1:100){ #again, 100 runs

  p1 <- rnorm(n, 0, sqrt(25)/2) #sigma A: 25
  p2 <- rnorm(n, 0, sqrt(25)/2)
  p3 <- rnorm(n, 0, sqrt(25)/2)
  p4 <- rnorm(n, 0, sqrt(25)/2)

  p1.pheno <- p1 + p2 + rnorm(n, 0, sigma.env)

  p2.pheno <- p3 + p4 + rnorm(n, 0, sigma.env)

  which.allele.p1 <- rbinom(n, 1,0.5)

  which.allele.p2 <- rbinom(n, 1, 0.5)

  o1.1 <- ifelse(which.allele.p1 == 0, p1, p2)


    o1.2 <- ifelse(which.allele.p2 == 0, p3, p4)

  o1.pheno <- o1.1+o1.2+rnorm(n, 0, sigma.env)

  VA.est[i] <- 2*cov(o1.pheno,p1.pheno)

  h2.est[i] <- 2* coef(lm(o1.pheno~p1.pheno))[2]
}

quantile(VA.est,c(0.025,0.975))
```

```
##     2.5%    97.5%
## 10.61854 14.35360
```

```r
quantile(h2.est, c(0.025,0.975))
```

```
##      2.5%     97.5%
## 0.8111712 1.0453391
```

```r
############## ##


sigma.env <- 15

for(i in 1:100){
```

```r
  p1 <- rnorm(n, 0, sqrt(25)/2)
  p2 <- rnorm(n, 0, sqrt(25)/2)
  p3 <- rnorm(n, 0, sqrt(25)/2)
  p4 <- rnorm(n, 0, sqrt(25)/2)

  p1.pheno <- p1 + p2 + rnorm(n, 0, sigma.env)

  p2.pheno <- p3 + p4 + rnorm(n, 0, sigma.env)

  which.allele.p1 <- rbinom(n, 1,0.5)

  which.allele.p2 <- rbinom(n, 1, 0.5)

  o1.1 <- ifelse(which.allele.p1 == 0, p1, p2)


    o1.2 <- ifelse(which.allele.p2 == 0, p3, p4)

  o1.pheno <- o1.1+o1.2+rnorm(n, 0, sigma.env)

  VA.est[i] <- 2*cov(o1.pheno,p1.pheno)

  h2.est[i] <- 2* coef(lm(o1.pheno~p1.pheno))[2]
}

quantile(VA.est,c(0.025,0.975))
```

```
##       2.5%     97.5%
## -12.64843  39.66722
```

```r
quantile(h2.est, c(0.025,0.975))
```

```
##         2.5%       97.5%
## -0.05204291  0.17106331
```

With sigma.env set to 15, our confidence intervals pass through 0, which indicates a lack of significance. The results are more favorable with sigma.env set to 1.

Part 3:

```r
h2.twin <- rep(0,100)
sigma.env <- 1

for(i in 1:100){

  p1 <- rnorm(n,0,sqrt(25)/2)
  p2 <- rnorm(n,0,sqrt(25)/2)
  p3 <- rnorm(n,0,sqrt(25)/2)
  p4 <- rnorm(n,0,sqrt(25)/2)
```

```r
  #dizygotic twins
  w1 <- rbinom(n,1,0.5)

  w2 <- rbinom(n,1,0.5)

  dz1 <- ifelse(w1 == 0,p1,p2)+ifelse(w2 == 0,p3,p4)+rnorm(n,0,sigma.env)

  w1 <- rbinom(n,1,0.5)

  w2 <- rbinom(n,1,0.5)

  dz2 <- ifelse(w1 == 0,p1,p2)+ifelse(w2 == 0,p3,p4)+rnorm(n,0,sigma.env)

  # monozygotic twins
  w1 <- rbinom(n,1,0.5)

  w2 <- rbinom(n,1,0.5)

  base <- ifelse(w1 == 0,p1,p2)+ifelse(w2 == 0,p3,p4)

  mz1 <- base+rnorm(n,0,sigma.env)

  mz2 <- base+rnorm(n,0,sigma.env)

  h2.twin[i] <- 2*(cor(mz1,mz2)-cor(dz1,dz2))

}

quantile(h2.twin,c(0.025,0.975))
```

```
##      2.5%     97.5%
## 0.827803 1.048726
```

```r
sigma.env <- 15

for(i in 1:100){

  p1 <- rnorm(n,0,sqrt(25)/2)
  p2 <- rnorm(n,0,sqrt(25)/2)
  p3 <- rnorm(n,0,sqrt(25)/2)
  p4 <- rnorm(n,0,sqrt(25)/2)

  #dizygotic twins
  w1 <- rbinom(n,1,0.5)

  w2 <- rbinom(n,1,0.5)

  dz1 <- ifelse(w1 == 0,p1,p2)+ifelse(w2 == 0,p3,p4)+rnorm(n,0,sigma.env)

  w1 <- rbinom(n,1, 0.5)

  w2 <- rbinom(n,1,0.5)
```

```r
  dz2 <- ifelse(w1 == 0,p1,p2)+ifelse(w2 == 0,p3,p4)+rnorm(n,0,sigma.env)

  # monozygotic twins

  w1 <- rbinom(n,1,0.5)

  w2 <- rbinom(n,1,0.5)

  base <- ifelse(w1 == 0,p1,p2) + ifelse(w2 == 0,p3,p4)

  mz1 <- base+rnorm(n,0,sigma.env)

  mz2 <- base+rnorm(n,0,sigma.env)

  h2.twin[i] <- 2*(cor(mz1,mz2)-cor(dz1,dz2))

}

quantile(h2.twin,c(0.025,0.975))
```

```
##       2.5%      97.5%
## -0.1385109  0.2561133
```

The confidence interval does not pass through 0 for sigma.env=1, but it does for sigma.env=15. It appears sigma.env = 1 is the preferable value.

Part 4:

Young, A. I. (2019). Solving the missing heritability problem. PLOS Genetics, 15(6), e1008222. https://doi.org/10.1371/journal.pgen.1008222

The author suggests improving heritability estimates using a methodology called Genomic Relatedness Restricted Maximum Likelihood, or GREML for short. With GREML, variance is estimated and used to measure the similarity of phenotype as it changes using SNPs as additional information.

To do this in R, we would need to simulate the SNPs for each individual as GREML would require more markers as we are only using one marker in this current example. We would also need the genotypes for each individual. Most importantly, we could use packages such as lme4 or GCTA to create the GREML model.